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Packet Accepted: 192.168.1.10 > 8.8.8.8
OUTBOUND PACKET ALERT: 203.0.113.45 on known malicious actor list! (10.0.0.5 > 203.0.113.45)
Packet Accepted: 172.16.0.22 > 1.1.1.1
INBOUND PACKET ALERT: 198.51.100.12 on known malicious actor list! (198.51.100.12 > 192.168.1.100)
Packet Accepted: 192.168.1.11 > 9.9.9.9
Packet Accepted: 192.168.1.12 > 64.233.177.99
INBOUND PACKET ALERT: 203.0.113.55 on known malicious actor list! (203.0.113.55 > 192.168.1.20)
Packet Accepted: 192.168.1.13 > 208.67.222.222
OUTBOUND PACKET ALERT: 192.0.2.123 on known malicious actor list! (192.168.1.14 > 192.0.2.123)
INBOUND PACKET ALERT: 198.51.100.200 on known malicious actor list! (198.51.100.200 > 192.168.1.15)
Packet Accepted: 192.168.1.16 > 52.34.22.8
Packet Accepted: 192.168.1.17 > 151.101.1.69
Packet Accepted: 192.168.1.18 > 104.16.249.249
Packet Accepted: 192.168.1.19 > 185.199.108.153
OUTBOUND PACKET ALERT: 198.51.100.77 on known malicious actor list! (10.0.0.25 > 198.51.100.77)
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IP-based IDS is a system that monitors network traffic, either on one single host, called HIDS, or on the network, called NIDS, and uses a known malicious actor list of IP addresses, alerting the IRT (Incident Response Team) or similar when it finds potentially malicious activity. Importantly, an IDS simply alerts the team – it does not filter or block traffic in any way. In many ways, a strictly IP-based IDS system is very similar to a firewall, using a series of rules based on IP addresses, however, as mentioned, in merely alerts when one of the rules hit.

This particular project clearly shows how this works, reading in a .txt document that lists a series of “known malicious IP addresses,” and then iterates through the “traffic” (a series of strings of IP addresses in an array), and verifies if they are on that list, alerting the user if they are. My project meets all requirements as laid out in the assignment sheet.

Similarly to the SQLi project (project 1), I also decided to go a step further, and create a whole application that can simulate an IDS system, including other potential indicators of compromise. It filters through each packet, sorts it into standard categories of “INFO”, “WARN”, “ALERT”, and “CRITICAL”, loading them into a tabbed table display, highlighting the critical packets. With these improvements, the application acts even more like a network-based IDS, even though it is still just a simulation. Of note – some of the classification values are tailored to the specific pcap file I used to provide some good outputs.